Natural History Museum (Grown-ups)

UK Family Tour

A Cathedral to Nature: Owen, Waterhouse and the Terracotta Zoo

The Natural History Museum is that rare thing, a building that argues a thesis. Richard Owen — the brilliant, prickly anatomist who coined the word Dinosauria in 1842 — spent twenty years campaigning to move natural history out of the British Museum's basement into a purpose-built "cathedral to nature," and Alfred Waterhouse delivered exactly that in 1881: a Romanesque basilica in blue and buff terracotta, chosen because it could be cast with ornament and would shrug off London's acid soot. The ornament is a program, not decoration: the west wing carries living species, the east wing extinct ones — a taxonomy in architecture — with pterodactyls on the roofline and monkeys scrambling up the arches of the central hall. There's an irony baked in: Owen was Darwin's most formidable scientific enemy, resisting natural selection while the museum was conceived, and the separation of living from extinct species reflects Owen's anti-evolutionary view that the two shouldn't be mixed. Darwin got the last word — his marble statue now presides over the main staircase, gazing down the hall, moved to that spot of honor in 2008 for his bicentenary, displacing... a statue of Owen. Museum politics have long memories. The collection behind the scenes is the real leviathan: eighty million specimens, including thousands collected by Darwin himself on the Beagle.

Reading the Terracotta and the Golden Ceiling

You already know the building tells a story, living animals carved on the west wing, extinct ones on the east. But let me give you the details that reward a slower look. Every one of those creatures is made of terracotta, baked clay, moulded and fired by a firm called Gibbs and Canning in Tamworth, then fixed to the building like a vast three-dimensional jigsaw. Waterhouse chose terracotta not only because London's filthy Victorian air ate ordinary stone, but because clay could be pressed into moulds and repeated, so an army of monkeys, wolves, pterodactyls, fish, and snarling wild cats could climb the arches and columns without bankrupting anyone. Walk slowly along the central hall and look up at the arch columns. The animals change as you go, and no guidebook has ever managed to name them all.

But the real showstopper is above your head, and most visitors never look up to see it. The ceiling of the great hall is not plain. It is divided into a hundred and sixty-two painted panels, glowing with gold leaf, and they are not decoration for their own sake. A hundred and eight of them are portraits of plants, each one chosen because it mattered to the museum, to Britain, or to the British Empire. There is tea and coffee and tobacco, cotton and sugar, the crops that built imperial fortunes, alongside medicinal herbs and English natives. They were painted by an artist named Charles James Lea, working from Waterhouse's designs, and the theme running through them is growth and power, nature as the engine of an empire. It is a botanical garden hanging over your head in gold, and it is one of the most beautiful and least noticed things in London.

Now, one darker chapter written into these very walls. On the ninth of September, nineteen forty, during the Blitz, bombs struck the east wing where the botany department kept its collections. Incendiaries and an oil bomb set the plant store alight, and priceless pressed specimens went up in flames. But there is a strange, almost miraculous footnote. As firefighters hosed down the wreckage, water soaked into ancient dried seeds that had been collected back in seventeen ninety-three, and some of them, after nearly a hundred and fifty years pressed flat in a cabinet, woke up and began to sprout among the ashes. Life, quite literally, coming back from the dead in the ruins of a bombed museum.

So before you rush to the whale, give the building three minutes. Look up at the terracotta beasts on the arches, then further up at the golden garden on the ceiling, and remember that both of them survived a war.

The Word Was Coined Here: Mantell, Owen and the Concrete Dinosaurs

The word dinosaur was coined by Richard Owen in eighteen forty-two, but the animals that let him coin it were mostly found by someone else, a country doctor named Gideon Mantell. Around eighteen twenty-two, Mantell, or possibly his wife Mary Ann, came across some enormous fossil teeth in a quarry in Sussex. Mantell puzzled over them, compared them to the teeth of a living iguana, and realised he was looking at a gigantic extinct plant-eating reptile. He named it Iguanodon, iguana-tooth, in eighteen twenty-five. It was only the second dinosaur ever named, and Mantell had opened a door.

Owen walked through it and, some would say, slammed it behind him. When he defined the group Dinosauria, he built it around Mantell's Iguanodon and two others, but he had a lifelong habit of quietly writing the discoverers out of the story and himself in. It was not a one-off. In eighteen forty-six Owen accepted the Royal Society's Royal Medal for a paper on a fossil, having failed to mention that an amateur named Chaning Pearce had actually discovered the creature years earlier. The scandal was serious enough that Owen was voted off the councils of both the Royal Society and the Zoological Society. He was a genius, and he was not always an honest one, and poor Mantell, injured in a carriage accident and endlessly overshadowed, got the worst of him.

Owen's fingerprints are on the world's very first dinosaur sculptures too. When the Crystal Palace was rebuilt in south London, the artist Benjamin Waterhouse Hawkins was commissioned to make life-sized models of these newly imagined monsters, under Owen's scientific direction, and they were unveiled in eighteen fifty-four. To celebrate, Hawkins held a now-legendary dinner for about twenty guests inside the hollow mould of an Iguanodon on New Year's Eve, with Owen enthroned at the head of the table. The models got the science charmingly wrong, four-legged, lumbering, with Iguanodon's thumb spike stuck on its nose as a horn, but they still stand in the park today, protected as national treasures, the ancestors of every dinosaur model since.

All of which makes the modern dinosaur gallery here a kind of family reunion. The animatronic Tyrannosaurus that roars and swings its head is the crowd-pleaser, but look also for Sophie the Stegosaurus, unveiled in twenty fourteen and the most complete Stegosaurus skeleton found anywhere in the world, over four-fifths real bone. From Mantell's handful of teeth in a Sussex quarry to a laser-scanned Stegosaurus and a robotic T. rex, this is where the whole idea of the dinosaur was born, named, quarrelled over, and, eventually, got mostly right.

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Hope and Dippy: What a Museum Chooses to Say

The 2017 decision to retire Dippy — the Diplodocus cast that had greeted visitors since 1979 — and suspend a real blue whale skeleton in Hintze Hall caused genuine national controversy, complete with a "Save Dippy" campaign. The museum's reasoning is worth knowing, because it's a small case study in what museums are for. Dippy was a plaster cast, a replica of an American fossil, gifted by Andrew Carnegie in 1905. The whale, by contrast, is real — a female that stranded at Wexford, Ireland, in 1891, bought by the museum for two hundred fifty pounds — and her story is an argument: blue whales were hunted to the brink, from perhaps a quarter million down to low thousands, until the 1966 international ban made them one of the first species humanity deliberately chose to save at global scale. Hence the name Hope, and the dynamic lunge-feeding pose, engineered from a hundred twenty-one bones on a bespoke armature. She's a statement that the museum's subject is no longer just cataloguing nature but negotiating our effect on it. Dippy, for the record, landed on his feet: after a triumphant national tour seen by millions, he's on long-term display in Coventry. The under-hull family sciencey bits aside, give the whale ninety seconds of pure looking — the skeleton's scale doesn't register on first glance; it accumulates.

The Treasures Gallery: A Feather, a Meteorite and a Piece of the Moon

If the whole museum had to be shrunk into one room, it would be the Treasures Gallery, formally the Cadogan Gallery, a dim jewel-box off the main hall where a couple of dozen objects are asked to stand for eighty million. It is worth a detour, because several of them are genuinely among the most important specimens on Earth.

Chief among them is a slab of pale limestone bearing the impression of Archaeopteryx, the so-called London specimen. Unearthed in Bavaria in eighteen sixty-one and bought by the museum the following year as part of a haul of Solnhofen fossils, it arrived at a perfect moment, just after Darwin published On the Origin of Species. Here was an animal caught exactly between two worlds, feathers and a wishbone like a bird, but teeth in its jaws, claws on its wings, and a long bony reptilian tail. It was, near enough, the missing link the Victorians were arguing about, evolution made visible in stone, and Richard Owen, of all people, secured it for the nation and described it. Whatever his feelings about Darwin, he knew treasure when he saw it.

A few cases away sits something that fell out of the sky.

The Wold Cottage meteorite crashed into a Yorkshire field in seventeen ninety-five, so close to a farmworker that it showered him with dirt. At the time, respectable science flatly denied that stones could fall from space. This one helped change that. It triggered the first serious investigation into meteorites, led in Britain by Sir Joseph Banks, and the conclusion, radical for its day, was that these rocks were genuinely extraterrestrial. You are looking at one of the stones that taught humanity the sky could drop things on us.

And then there is the Moon. The gallery holds the United Kingdom's Goodwill Moon rock, a fragment brought back by the final Apollo mission and presented to Britain, mounted beside a small Union flag that was itself carried to the lunar surface and home again. It is the size of a thumbnail and it is a piece of another world, picked up by human hands.

Scattered among these are specimens collected by Darwin himself on the voyage of the Beagle. That is the argument the room is quietly making. A great natural history museum is not a warehouse of dead animals but a library of evidence, the feather that proved birds descend from dinosaurs, the stone that proved space rains rock, the pebble that proved humans walked on the Moon. Give this small room more time than its size suggests, because it holds the museum's whole thesis in miniature.

The Whale's Darker Arithmetic

Hope's story, as the museum tells it, is a rescue, and it is true. Blue whales were hunted to the very edge, humanity chose to stop, and the numbers are creeping back. But there is a darker chapter behind the triumph, and it is worth knowing, because it complicates the tidy tale of the nineteen sixty-six ban in a very human way.

For decades everyone assumed that once the international rules were agreed, the killing largely stopped. It did not. In nineteen ninety-three, a Russian scientist named Alexey Yablokov, who had once sailed with the Soviet whaling fleets, revealed that the Soviet Union had spent decades systematically cheating. Its factory ships kept two sets of logbooks, a false one for the international inspectors and a true one for Moscow, and between the late nineteen forties and the early seventies they secretly slaughtered something like a hundred and eighty thousand whales they never reported, including protected blue whales, mothers, and calves that the rules expressly forbade them to touch.

The most chilling part is how pointless much of it was. The Soviet fleets had production quotas to meet, tonnages set by planners far from the sea, so they killed whales they could not even process, sometimes taking only a fraction of the animal and leaving the rest to rot. In the Southern Ocean alone, across the whole grim century of industrial whaling, roughly a third of a million whales were killed while barely more than half that number were ever declared. A generation of scientists studying whale recovery had been working from figures that were quietly fictional.

Why tell you this beneath a skeleton named Hope? Because it sharpens what she means. The recovery of the blue whale is real, and it is genuinely one of the first times our species deliberately pulled a giant back from the brink at a global scale. But it was nearly undone in secret, by a state that signed the treaties and broke them in the dark, and we only know the true scale because a few insiders kept honest records and, eventually, told the truth.

So when you stand under Hope, hold both halves of the story. She is an emblem of what conservation can achieve, the largest animal that has ever lived, lunging over your head instead of vanishing into history. And she is a reminder that agreements on paper are only as good as the honesty of the people keeping the books, and that the real work of saving anything is slow, unglamorous, and easy to cheat. Hope is exactly the right name. But it was, and still is, a close-run thing.

The Shaking Floor and the Vault

Two experiences in the museum's quieter wings are worth planning for, and they could not be more different, one all violence, one all serenity. First, in the Volcanoes and Earthquakes gallery, find the earthquake simulator, a small mock-up of a Japanese convenience store. Step onto the platform, hold the rail, and the floor recreates the Great Hanshin earthquake that devastated the city of Kobe on the seventeenth of January, nineteen ninety-five, a quake of around magnitude seven that killed more than six thousand people. The shelves judder, the goods rattle, the lights fail, and a screen plays genuine security-camera footage recorded in a Kobe shop as it happened. It lasts only seconds, and it is deliberately restrained, but it conveys something no diagram can, the specific horror of the one thing you always trusted, the ground itself, betraying you.

It belongs beside the volcanoes for a reason. Both are products of plate tectonics, the slow churning of the Earth's fractured crust. Where the great plates grind and catch and suddenly slip, the ground quakes, and where they pull apart or dive beneath one another, molten rock finds its way up. The gallery is really a portrait of a restless planet, and the shaking supermarket is its most honest exhibit, because it makes you feel, for a moment, genuinely unsafe.

Then, for the opposite of terror, climb to the Vault, the museum's treasury of gemstones and minerals, and find the Aurora Pyramid of Hope. It is a stepped display of two hundred and ninety-six naturally coloured diamonds, the most comprehensive collection of natural-colour diamonds anywhere in the world, arranged in a glittering pyramid of yellows, greens, pinks, blues, and rare reds. It came here from New York in two thousand and five. Linger, because it does two tricks. Under ultraviolet light, some of the stones fluoresce in unearthly colours, glowing where a moment before they merely sparkled. And at the very apex sits a small diamond that will actually change colour, drifting from olive-green to yellow depending on the light it has lately been keeping.

The Vault around it holds some of the rarest objects the Earth produces, gem-quality crystals, a piece of Mars, extraordinary nuggets, the sort of things usually locked in bank vaults or royal collections. Taken together with the shaking floor two galleries away, they make a pairing worth seeking out, the planet at its most destructive and the planet at its most exquisite. The same geology that flattens cities also spends millions of years growing a single flawless coloured diamond. It is all the same restless Earth, and this museum keeps both of its faces.

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Mary Anning and the Museum's Hidden Nine-Tenths

The fossil sea reptiles along the gallery walls carry a story the labels used to omit. Mary Anning, born 1799 in Lyme Regis to a poor cabinetmaker's family, was perhaps the greatest fossil hunter who ever lived: with her brother she excavated the first scientifically described ichthyosaur when she was around twelve, went on to find the first complete plesiosaur — so strange that Cuvier in Paris initially suspected a fake — and the first pterosaur outside Germany. Gentleman geologists of the Geological Society, which she could not join as a woman and would not have joined as working-class, bought her specimens and published on them, credit optional. She died at forty-seven; the Society, to its partial credit, had raised money for her in her final illness, and today several of her actual finds hang here, properly attributed. She's also plausibly the subject of the tongue-twister "she sells seashells." Beyond the galleries, remember that the display is the museum's tip: some eighty million specimens sit behind the scenes, from Darwin's Beagle collections to frozen tissue banks, and around three hundred scientists work here — it remains one of the world's most active taxonomic research institutions, describing hundreds of new species a year. The gift shop dinosaurs are, admittedly, also world-class.

The Sea Dragons Themselves

We celebrate Mary Anning for finding these creatures, but pause on the animals themselves, because they are far stranger than the word fossil suggests, and they overturned how people understood the world. Start with the ichthyosaur, the fish-lizard. It looks, at a glance, like a dolphin, streamlined body, pointed snout, a fin on its back and a crescent tail. But it was a reptile, a cousin of lizards and snakes that abandoned the land entirely and went back to the sea, breathing air, ruling the oceans while the dinosaurs ruled the shore. Its resemblance to a dolphin is one of the great coincidences of evolution, two utterly unrelated animals, tens of millions of years apart, shaped by the same problem, moving fast through water, into nearly the same body.

Ichthyosaurs had enormous eyes, among the largest of any animal that has ever lived, some the size of dinner plates, for hunting in the dark of the deep. And unlike most reptiles, they did not lay eggs on a beach. They gave birth to live young in open water, and we know this because fossils have been found of mothers frozen in the very act, a baby emerging tail-first, exactly as a whale is born. To hold one of Anning's ichthyosaurs is to look at a reptile that had reinvented itself as a fish and a whale at once.

The plesiosaur is stranger still, and you have seen its silhouette without knowing it, because it is the shape every artist reaches for when they draw the Loch Ness monster, a broad turtle-like body, four great wing-shaped flippers, and, in many species, an impossibly long neck. When Anning found the first complete one, its neck had so many more bones than any known reptile that the great anatomist Cuvier assumed it had to be a forgery, until he saw that it was real. Plesiosaurs flew through the water with those four flippers, something no animal alive does today, and no one is entirely sure how.

Here is why these animals mattered beyond their oddity. In Anning's lifetime, many people still believed that no species could ever truly vanish, that a wise Creation would not discard its own creatures. Yet here were monsters preserved in the cliffs, plainly real, plainly nowhere in the living oceans. The sea dragons were physical proof that whole kinds of life had existed and then ended, that the history of the Earth was vastly longer and stranger than scripture allowed. Mary Anning, hammering along a crumbling Dorset shore, was not merely collecting curiosities. She was pulling up the evidence that extinction is real, and quietly helping to remake the human understanding of time itself.

Dorothea Bate and the Women of the Museum

Mary Anning was kept outside the door of science by her sex and her class. It is worth knowing that the very museum now displaying her fossils under her own name took a long time to open that door too, and that the woman who forced it deserves to be far more famous than she is. Her name was Dorothea Bate.

In eighteen ninety-eight, aged nineteen and with almost no formal schooling, Bate simply turned up at this museum and talked her way into a job in the bird room, sorting and cleaning bird skins. She was, as far as anyone can tell, the first woman employed here as a scientist, and for a long time she was not paid a proper salary at all, but by the piece, so much for each fossil she prepared. She stayed for fifty years, teaching herself geology, anatomy, and palaeontology as she went, exactly as Anning had taught herself a century before.

And then she did something remarkable. In the first decade of the twentieth century she sailed, largely at her own expense, to the islands of the Mediterranean, to Cyprus, Crete, and the Balearics, and went hunting through their caves. What she pulled out rewrote a chapter of natural history. She found elephants the size of ponies and hippos no bigger than pigs, dwarf species that had evolved in miniature because they were marooned on small islands with little food and few predators. Bate had uncovered island dwarfism, one of the strangest and most reliable tricks in evolution, whereby giants shrink and, elsewhere, small creatures balloon, simply because an island rewrites the rules. Much of what we understand about how isolation reshapes animals begins with her digging in those caves.

She kept working almost to the end of her long life, identifying, late in her career, the first fossil elephant known from the Holy Land. For decades she was a quiet fixture of this institution, respected by the scientists around her, and yet for a long while barely known to the public, her name absent from the galleries her work helped to fill.

So as you look at Anning's sea dragons, hold the two women together in your mind. One worked the cliffs of Dorset and was written out of the papers by the gentlemen who bought her finds. The other worked the caves of the Mediterranean and the back rooms of this very building, paid by the specimen, and made herself indispensable. The museum has lately begun to say their names aloud, with a blue plaque here for Bate and a hard-won statue for Anning in Lyme Regis. It is late, and it is right, and it is the same long story slowly bending toward credit.